Product Requirements Document

Hospital Multi-Model AI Diagnostic & Cost Prediction System

Graduation Project (AI / Data Science)

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1. Executive Summary

This project is a web-based, authenticated hospital analytics system built with Streamlit. After logging in, the user lands on a dashboard containing two independent tabs, each wrapping a separate machine learning model:

- Tab 1 — Insurance Cost Prediction (Regression): predicts the expected medical insurance charge for a patient based on demographic and lifestyle attributes.
- Tab 2 — Chest Disease Detection (Multi-Class Image Classification / Object Detection): takes a chest X-ray image as input and predicts the disease class present, localizing the abnormal region with a bounding box.

The system is designed as a single unified portal so hospital/insurance staff can use one login to access two distinct decision-support tools instead of two separate applications.

2. Problem Statement

Hospitals and insurance providers currently rely on manual, disconnected processes for two recurring tasks:

- Estimating insurance cost for a patient is done manually using flat actuarial tables that don't account for the interaction between variables (e.g., a smoker with high BMI costs disproportionately more than the sum of each factor alone).
- Reading chest X-rays for disease screening depends entirely on radiologist availability, which creates bottlenecks, delays, and is prone to human fatigue-related error, especially in high-volume or under-staffed facilities.

There is no single internal tool that gives staff quick, data-driven, model-backed answers to both problems in one place.

3. Goals and Objectives

#	Objective	Success Indicator
1	Provide instant, model-based insurance cost estimation	Regression model achieves acceptable R^2 / low RMSE on test data
2	Provide instant, model-based chest disease screening from X-rays	Classification/detection model achieves acceptable accuracy / mAP
3	Restrict access to authorized hospital staff only	Login gate blocks unauthenticated access
4	Present both tools in one consistent, simple interface	Single Streamlit app, two tabs, consistent UX

#	Objective	Success Indicator
5	Make output explainable enough for non-technical staff to trust it	Result screen shows prediction + supporting context

Out of Scope

- Real diagnosis / prescribing treatment — this is a decision-support tool, not a replacement for a doctor's diagnosis.
- Multi-hospital, multi-tenant architecture (single-instance deployment only).
- Mobile native app (web-responsive Streamlit only).
- Integration with real hospital EMR/HIS systems (data is entered manually or uploaded per session).

4. Target Users / Stakeholders

User Type	Role in the System
Hospital administrative / insurance staff	Uses Tab 1 to estimate a patient's insurance charge during intake or policy review
Radiology technician / triage nurse	Uses Tab 2 to get a first-pass screening read on a chest X-ray before a radiologist reviews it
System Admin (project supervisor / IT)	Manages user accounts, monitors system usage
Graduation Project Evaluator	Reviews the system as an academic deliverable (data science + engineering rigor)

5. High-Level System Flow

1. Landing Page → Login Screen — the user must authenticate with a username and password before reaching any functionality. No tab, model, or data is accessible pre-login.

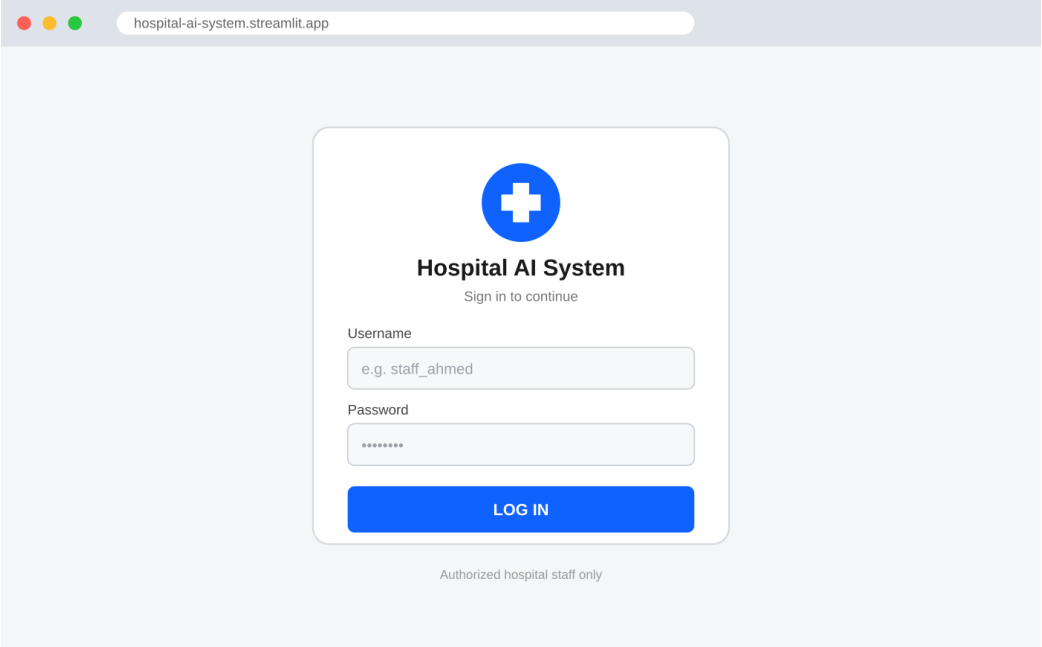


Figure 1 — Login screen wireframe (sample)

2. Post-Login → Dashboard with Two Tabs — once authenticated, the user sees a single dashboard page with two tabs (Streamlit st.tabs()): "Insurance Cost Prediction" and "Chest Disease Detection". Each tab is a self-contained mini-application with its own input form, its own Predict/Analyze button, and its own result panel. Switching tabs does not reset or affect the other tab's state.

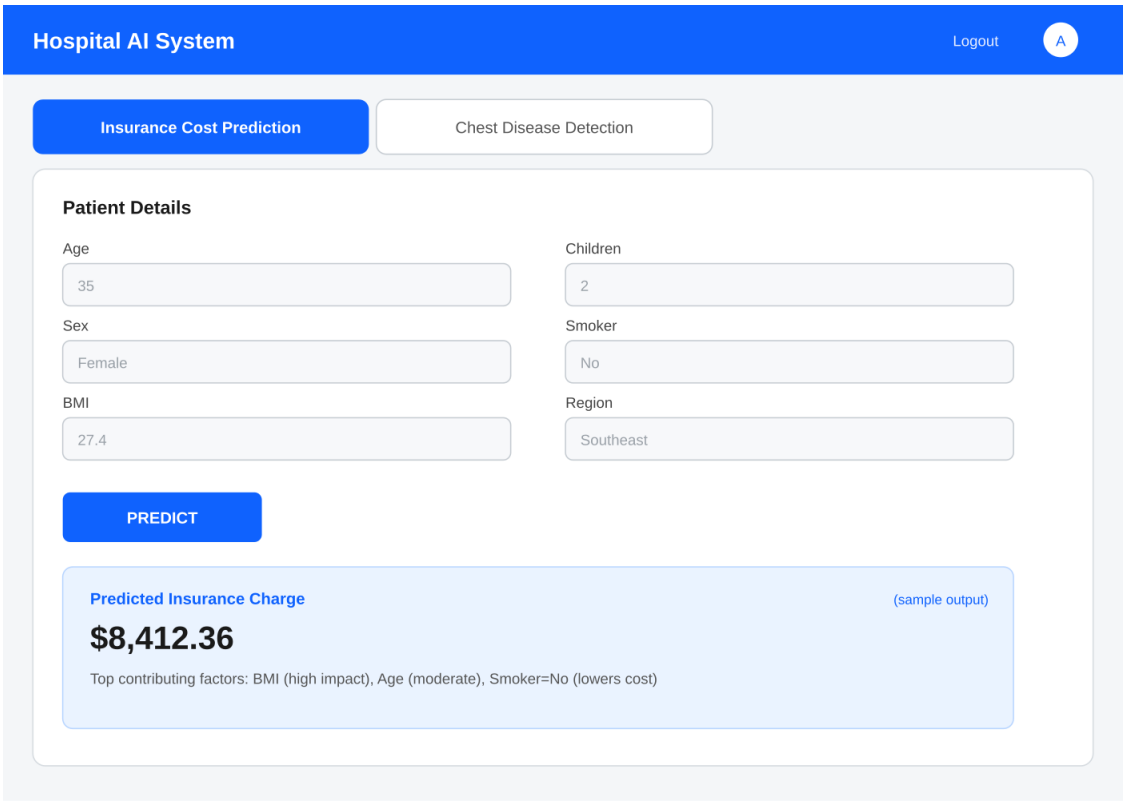


Figure 2 — Dashboard, Tab 1 (Insurance Cost Prediction) with a sample prediction

3. Tab 1 Interaction Flow — user fills a form (age, sex, BMI, children, smoker status, region) → clicks Predict → system runs the trained regression model → displays the predicted insurance charge with supporting context (e.g., which factors pushed the cost up/down).

4. Tab 2 Interaction Flow — user uploads a chest X-ray image (jpg/png/dicom) → clicks Analyze → system runs the trained detection/classification model → displays the original image with a bounding box drawn around the detected abnormality, the predicted disease class, and a confidence score.

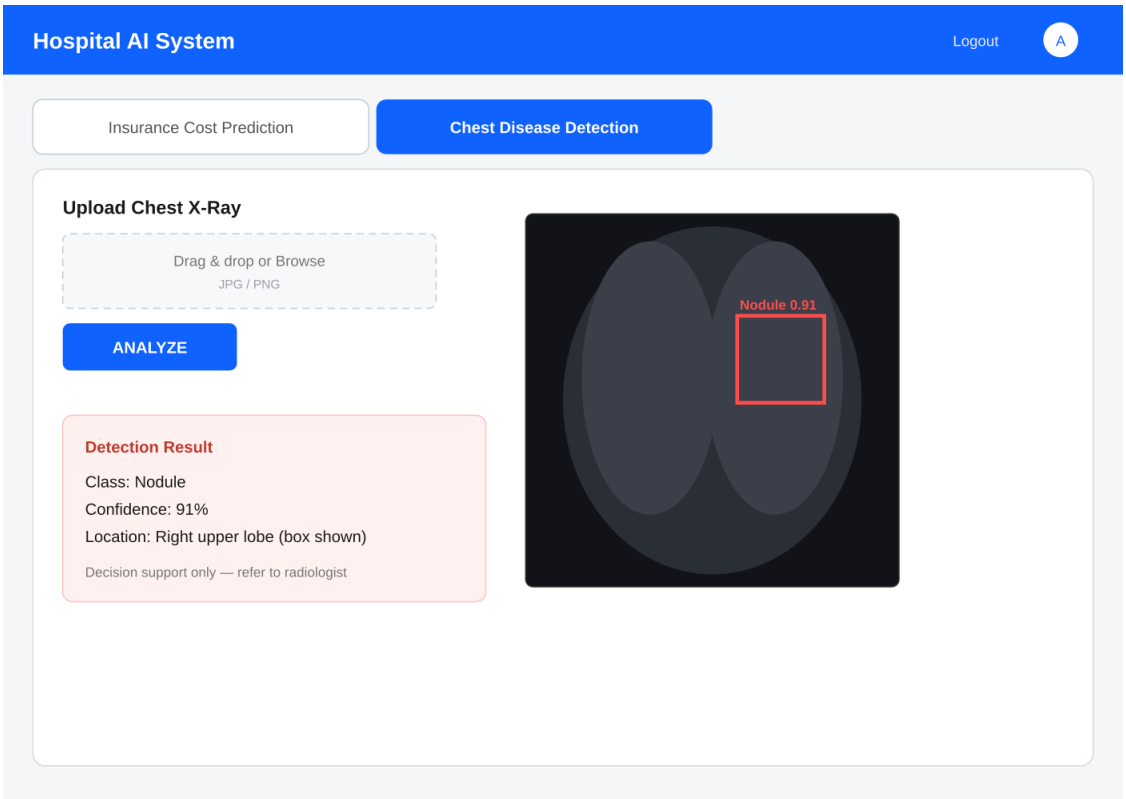


Figure 3 — Dashboard, Tab 2 (Chest Disease Detection) with a sample detection result

6. Business Understanding of the Data (Required Before Any Modeling)

A graduation committee will specifically look for this section — it proves you understand why the data looks the way it does and what business problem it encodes, not just that you can call .fit(). Cover both datasets like this in your actual project report:

6.1 Dataset 1 — Medical Insurance Cost Prediction (Regression)

Source: Kaggle — Medical Insurance Cost Prediction

What this data represents (business context)

This dataset simulates how a health insurance company calculates the premium/charge it bills a policyholder. In the real world, insurers use actuarial models — this dataset is a simplified, tabular version of that logic.

Typical columns and their business meaning

Column	Business Meaning
age	Older patients statistically incur higher medical costs → higher premium
sex	Used to check for/avoid biased pricing; some medical risk factors differ by sex
bmi	Body Mass Index — a proxy for obesity-related health risk (diabetes, heart disease)
children	Number of dependents covered under the policy — affects total covered cost
smoker	Historically the single strongest cost driver — smokers have dramatically higher claims
region	Cost of healthcare/living varies geographically
charges (target)	The actual dollar amount billed — what the regression model must learn to predict

Business questions this model answers

- "If a new applicant walks in with these attributes, what should we expect to charge them?"
- "Which factor (age, smoking, BMI) contributes most to driving cost up?" — useful for wellness-program justification (e.g., quitting smoking lowers risk-adjusted cost).

Why regression

The target (charges) is a continuous numeric value, not a category — this is a supervised regression problem, not classification.

Data-prep considerations to document in the report

- Encoding categorical fields (sex, smoker, region).
- Checking for outliers in charges — smokers create a bimodal cost distribution.
- Checking multicollinearity between features.
- Scaling numeric features depending on the algorithm chosen (not required for tree-based models; required for linear/regularized regression or distance-based models).

6.2 Dataset 2 — Chest X-Ray Disease Detection (Multi-Class + Bounding Boxes)

Source: Kaggle — CheXDet Image and Annotations

What this data represents (business context)

This is a medical imaging screening dataset: chest X-rays paired with radiologist-style annotations that mark where on the image a disease finding is located (bounding box) and what it is (the disease class). This mirrors the real radiology workflow: a radiologist looks at a film, spots an abnormal region, circles/boxes it, and names the finding in their report.

Structure to document precisely in your report

Verify exact values from the actual annotation file, since label sets vary between chest X-ray datasets:

- Images: raw chest X-ray films (typically grayscale, PA/AP view).
- Annotations: bounding box coordinates (x, y, width, height or x_min/y_min/x_max/y_max) per finding, plus a class label naming the pathology (common chest X-ray finding categories include Atelectasis, Cardiomegaly, Effusion, Nodule/Mass, Pneumothorax, Consolidation, Fibrosis — confirm the exact class list against the dataset's own annotation CSV/JSON before finalizing your report).
- A single image may contain zero, one, or multiple annotated findings.

Business questions this model answers

- "Does this X-ray show a visible abnormality, and if so, what type and where?"
- Used as a first-pass triage filter: flag likely-abnormal scans for priority radiologist review, and let clearly normal scans move faster through the queue.

Why multi-class classification + object detection

The business need isn't just "sick / not sick" — staff need to know which condition is suspected and where it is, which is what justifies the added complexity of bounding-box object detection (e.g., YOLO, Faster R-CNN) over plain image classification (e.g., a CNN classifier).

Data-prep considerations to document in the report

- Image resizing/normalization.
- Class imbalance — some diseases are rarer than others; may require augmentation or weighted loss.
- Converting annotation format to what your chosen detection framework expects (YOLO .txt vs. COCO JSON vs. Pascal VOC XML).

- Train/val/test split done by patient, not by image, if a patient has multiple images (to avoid data leakage).

7. Detailed System Architecture

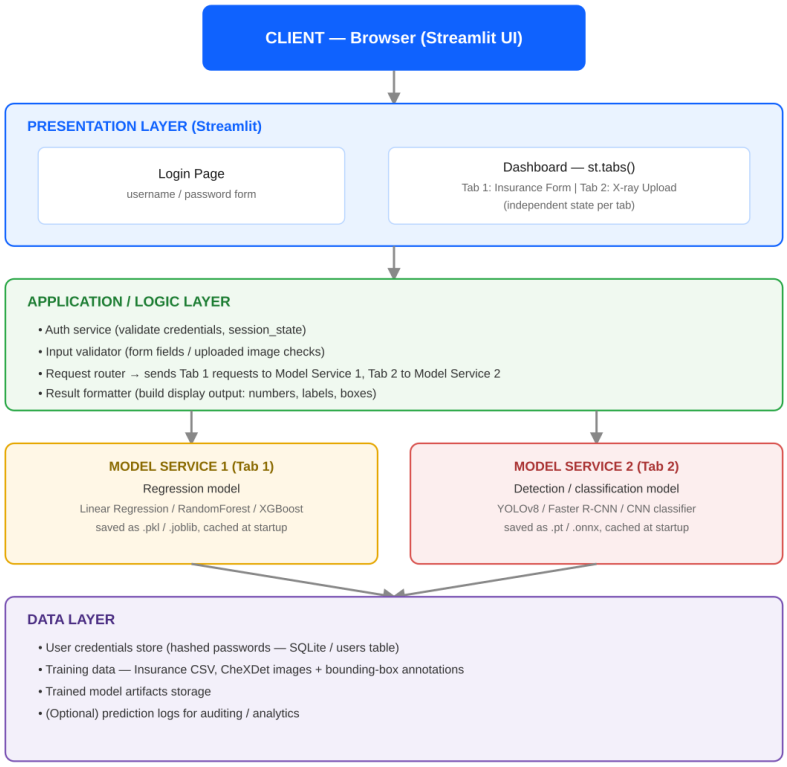


Figure 4 — Layered system architecture (sample)

Layer responsibilities

Layer	Responsibility
Presentation	Renders login form, tabs, input widgets, and results using Streamlit components
Application/Logic	Authenticates users, validates inputs, routes requests to the correct model, formats output
Model Service 1	Loads the pre-trained regression model once (cached) and returns a predicted charge
Model Service 2	Loads the pre-trained detection/classification model once (cached) and returns class + bounding box + confidence
Data Layer	Stores user credentials, raw datasets, trained model artifacts, and optional prediction logs

8. Detailed Functional Requirements

8.1 Authentication Module

- FR-1.1: System shall present a login form (username, password) as the first screen.
- FR-1.2: System shall reject access to the dashboard for unauthenticated sessions.
- FR-1.3: Passwords shall be stored hashed, never in plain text.
- FR-1.4: System shall support session persistence via Streamlit session_state so the user isn't logged out when switching tabs.
- FR-1.5: System shall provide a logout action.

8.2 Dashboard / Navigation

- FR-2.1: After login, system shall display a two-tab interface using st.tabs(["Insurance Cost Prediction", "Chest Disease Detection"]).
- FR-2.2: Each tab shall retain its own input/output state independently.

8.3 Tab 1 — Insurance Cost Prediction

- FR-3.1: System shall present input fields for: age, sex, BMI, number of children, smoker status, region.
- FR-3.2: System shall validate inputs (e.g., age > 0, BMI within realistic bounds) before allowing prediction.
- FR-3.3: On clicking "Predict", system shall pass inputs to the trained regression model and display the predicted charge.
- FR-3.4: System shall display a short explanation/breakdown of which inputs most influenced the predicted charge (e.g., via feature importance or SHAP values).

8.4 Tab 2 — Chest Disease Detection

- FR-4.1: System shall allow the user to upload a chest X-ray image (jpg/png).
- FR-4.2: System shall validate the uploaded file is a supported image type/size before processing.
- FR-4.3: On clicking "Analyze", system shall run the image through the trained detection model.
- FR-4.4: System shall display the image with bounding box(es) drawn over detected finding(s).
- FR-4.5: System shall display the predicted class label(s) and confidence score(s) alongside the image.
- FR-4.6: If no abnormality is detected, system shall clearly state "No abnormal finding detected" rather than leaving the result ambiguous.

9. Non-Functional Requirements

Category	Requirement
Performance	Regression prediction should return in <1s; image inference should return in <5s on standard hardware
Security	Passwords hashed; no raw patient-identifying data required beyond what's needed for prediction
Usability	Non-technical hospital staff should be able to use both tabs without training
Reliability	Models load once at app startup (cached) rather than reloading per request
Portability	App should run via streamlit run app.py with a documented requirements.txt
Explainability	Every prediction should be shown with some form of supporting context, not a bare number/label

10. ML Pipeline per Tab

For each tab, structure your work using this sequence (standard CRISP-DM applied to your case):

- Business Understanding — as documented in Section 6.
- Data Understanding — EDA: distributions, class balance, correlations (Tab 1), sample image inspection and annotation density (Tab 2).
- Data Preparation — cleaning, encoding, splitting, augmentation (Tab 2 only), image resizing/normalization.
- Modeling — Tab 1: try 2–3 regression algorithms (Linear Regression baseline, Random Forest, XGBoost) and compare. Tab 2: choose a detection architecture (YOLOv8 is fastest to prototype) or a two-stage classifier+localizer approach.
- Evaluation — Tab 1: RMSE, MAE, R^2. Tab 2: mAP (mean Average Precision) for detection, accuracy/F1 per class for classification.
- Deployment — export the best model (.pkl/.joblib for Tab 1, .pt/.onnx for Tab 2) and load it inside the Streamlit app.

11. Suggested Tech Stack

Component	Suggested Tool
UI Framework	Streamlit
Regression Modeling	scikit-learn / XGBoost
Image Modeling	PyTorch or Ultralytics YOLOv8, OpenCV for drawing boxes
Auth	streamlit-authenticator package or a simple hashed-password table (SQLite)
Data Storage	SQLite or CSV files (sufficient for a graduation project)
Explainability	SHAP (Tab 1)
Deployment	Streamlit Community Cloud or a local/university server

12. Risks & Assumptions

Risk / Assumption	Mitigation
CheXDet class labels/annotation format unconfirmed until inspected directly	Inspect the actual annotation file first; document exact classes before finalizing modeling section
Image dataset may have class imbalance	Use augmentation and/or class-weighted loss
This is a decision-support tool, not a certified diagnostic device	State this clearly as a disclaimer in the UI and report
Small dataset size limits deep model performance	Consider transfer learning (pretrained CNN/YOLO backbones) rather than training from scratch

13. Success Metrics (Definition of Done)

☐ Functional login gate blocking unauthenticated access

☐ Tab 1 returns a plausible predicted charge with model evaluation metrics documented (RMSE/R^2)

☐ Tab 2 returns a class label + bounding box + confidence with detection metrics documented (mAP)

☐ Business Understanding section (Section 6 style) present in the final academic report

☐ End-to-end demo runs without crashing on a sample login → Tab 1 prediction → Tab 2 prediction walkthrough

14. Appendix — Data Sources

Dataset 1: Medical Insurance Cost Prediction

Dataset 2: CheXDet Image and Annotations